AI&ML (ARTIFICIAL INTELLIGENCE AND MECHINE LEARNING)

AN INTERNSHIP TRAINING REPORT

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF COMPUTER APPLICATIONS

to the

Thiruvalluvar University, Serkadu, Vellore – 632 115.

by

HAMATHI THAZIN.A (34624U09028)

Under the guidance of

Mrs.N.NITHYALAKSHMI, M.Sc., M.Phil., SET,(Ph.D).,

HEAD,DEPARTMENT OF COMPUTER APPLICATIONS

M.M.E.S WOMEN'S ARTS AND SCIENCE COLLEGE

Hakeem Nagar,Melvisharam-632 509

OCTOBER – 2026

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STRYDO
TECHNOLOGIES PVT. LTD.

CERTIFICATE OF INTERNSHIP

PROUDLY PRESENTED TO

Hamathi Thazin A

A **BCA** Student from **M.M.E.S WOMEN'S ARTS AND SCIENCE COLLEGE** has successfully completed an internship training in **ARTIFICIAL INTELLIGENCE& MACHINE LEARNING** from **21th MAY** to **8th JUNE.**

Strydo Technologies Pvt. Ltd. Vellore

Founder & CEO

08-06-2026

Date of Issue

Certificate ID:202613159 .

BONAFIDE CERTIFICATE

This is to certify that Mr/Ms Hamathi Thazin. A of College/Institution worked as an intern as part of her B.C.A course in Computer Applications of Thiruvalluvar University. The particulars of internship are given below:

Internship starting date: 21/05/2026

Internship ending date: 08/06/2026

Actual number of days worked: 15 days

Tentative number of hours worked: 15 hours

Hours Broad area of work:

A small description of work done by the intern during the period:

during the period of time, she was working good and punctual.

Signature: A. Dilfar Nisha

Name: A. Dilfar Nisha

Designation: Data Scientist

Contact Number: 9362620620

Email: vellore.strydo@gmail.com

Stryde Technologies Pvt. Ltd. Vellore - 632 007

(Seal of the organization)

Professional Evaluation of Intern

Name of Intern:

College/Institution:

[Note: Give a score in the 1-5 scale by putting √ in the respective cells]

S. No	Particular	Excellent	Very Good	Good	Moderate	Satisfactory
1	Attendance		✓			
2	Punctuality		✓			
3	Adaptability	✓				
4	Ability to shoulder responsibility		✓			
5	Ability to work in a team	✓				
6	Written and oral communication skills		✓			
7	Problem solving skills		✓			
8	Ability to grasp new concepts		✓			
9	Ability to complete task	✓				
10	Quality of work done			✓		

Comments:

During the period of her internship with us, she was good.

Signature: A. Dilfar Nisha

Name: A. DILFAR NISHA

Designation: Data scientist

Contact Number: 9362620620

Email: vellore.strydo@gmail.com

Strydo Technologies Pvt. Ltd. Vellore - 632 007

(Seal of the organization)

CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that the internship work on **"AI&ML (ARTIFICIAL INTELLIGENCE AND MECHINE LEARNING)"** is the Bonafide work carried out by **HAMATHI THAZIN A (34624U09028)**, of III-BCA **M.M.E.S WOMEN'S ARTS AND SCIENCE COLLEGE** submitted to **THIRUVALLUVAR UNIVERSITY.** This is an authentic record of their Internship work carried out during their IV semester vacation under the Guidance of **Mrs. N. Nithyalakshmi, M.Sc., M.Phil., SET., (Ph.D.).,** Head, of the Department of Computer Application.

10/2/26

SIGNATURE OF THE
HEAD OF THE DEPARTMENT

Head Of the Department of
Computer Applications,
M.M.E.S. Women's
Arts & Science College,
Hakeem Nagar,
Melvisharam-632509

ACKNOWLEDGEMENT

First of all, I wholeheartedly thank the Almighty for having given me a good opportunity andstrength to successfully complete my internship.

I would like to extend my sincere thanks to our esteemed Secretary/Correspondent, **Dr. K. Anees Ahmed Sahib, B.A., D.Lit.** for his valuable guidance throughout my internship.

I forward my respectful thanks to our beloved and caring Principal, **Dr. Freda Gnanaselvam, M.A., M.B.A., M.M.M., Ph.D.,** who allowed me to undergo the internship.

I express my sincere gratitude to our Head of the Department, **Mrs. N. Nithyalakshmi, M.Sc., M.Phil., SET.,(Ph.D).,** for giving the valuable guidance and advice, patiently, for time to time, without which the internship could not have been a success.

I take great privilege in expressing our heartfelt and sincere gratitude to our internship guide **Mrs. N. Nithyalakshmi, M.Sc., M.Phil., SET.,(Ph.D).,** for providing valuable suggestions and motivation throughout the internship and made me complete the internship on time.

I sincerely thank **STRYDO TECHNOLOGIES** for giving me the opportunity and support to complete my internship.

I am grateful to my guide, **Ms. DILFAR NISHA A., (Data Science)** , by her valuable guidance throughout my internship.

ABSTRACT

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The internship at Strydo Technologies Pvt. Ltd. provided an excellent opportunity to gain practical exposure to Artificial Intelligence and Machine Learning technologies. During the fifteen-day internship, I learned Python programming, data structures, object-oriented programming, data preprocessing, machine learning algorithms, neural networks, deep learning, model evaluation, and ethical aspects of Artificial Intelligence.

The internship combined theoretical knowledge with practical implementation through hands-on exercises and real-world examples. Various Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, and Keras were introduced for data analysis and machine learning model development.

This internship enhanced my technical skills, analytical thinking, programming knowledge, and problem-solving abilities. It also improved my confidence in working with AI technologies and provided valuable industry exposure that will help me pursue a successful career in Artificial Intelligence and Machine Learning.

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CHAPTER 1
COMPANY PROFILE

1. COMPANY PROFILE

1.1 Company History

Strydo Technologies Pvt. Ltd. is an **ISO 9001:2015 certified company** and a part of the **STRYDO Group of Companies**. It is one of India's leading organizations in Information Technology training, research, development, and project-based learning. The company was established with the vision of bridging the gap between academic education and industrial requirements by providing practical knowledge and hands-on experience to students and professionals.

The organization focuses on delivering quality training programs in various technologies while encouraging innovation, research, and skill development. Over the years, Strydo Technologies has trained thousands of students in emerging technologies such as Artificial Intelligence, Machine Learning, Data Science, Embedded Systems, Mobile Application Development, and Web Technologies.

The company has earned a strong reputation by providing industry-oriented training, customized software development, hardware integration solutions, and research support.Through continuous innovation and commitment to quality, Strydo Technologies has become one of the preferred destinations for technical internships and industrial training.

1.2 Vision and Mission

Vision

To become a globally recognized organization that empowers students and professionals with industry-ready technical skills through innovation, quality education, and practical learning.

Mission

- To provide quality technical education through practical learning.

- To bridge the gap between academics and industry.
- To encourage innovation and research among students.
- To deliver customer-oriented software and hardware solutions.
- To produce highly skilled professionals capable of solving real-world problems.

1.3 Services Offered

Strydo Technologies provides a wide range of services across various technology domains.

Major Services

- Mobile Application Development
- Artificial Intelligence Solutions
- Machine Learning Solutions
- Research and Development
- MATLAB Projects
- Communication Systems
- Industrial Projects
- Cloud Computing Solutions

The company maintains a highly skilled team of professionals dedicated to providing quality solutions while ensuring customer satisfaction through innovation, transparency, and technical.

CHAPTER 2
ACTIVITIES OF VARIOUS DEPARTMENT

2. ACTIVITIES OF VARIOUS DEPARTMENTS

2.1 Activities

The AI & ML department is responsible for designing intelligent systems capable of learning from data and making predictions without explicit programming. The department focuses on research, software development, data analysis, and machine learning model deployment.

The development process generally includes:

- Problem identification
- Data collection
- Data preprocessing
- Feature engineering
- Model selection
- Model training
- Model evaluation
- Model optimization
- Model deployment
- Performance monitoring

The department works extensively with programming languages like **Python** and utilizes variouslibraries including NumPy, Pandas, Matplotlib, Seaborn, TensorFlow, Keras, and Scikit.

Modern AI development also requires knowledge of cloud platforms, version control systems, Docker, Kubernetes, MLflow, APIs, and deployment pipelines to build scalable machine learning solutions.

CHAPTER 3
JOB DESCRIPTION

3. JOB DESCRIPTION

3.1 INTERNSHIP EXPERIENCE

The internship was conducted for **15 days**, during which I received practical training in Python programming, Machine Learning concepts, Artificial Intelligence applications, Deep Learning, and Model Development.Each day focused on different concepts that gradually built my understanding from programming fundamentals to advanced AI applications.

3.2 INTERNSHIP EXPERIENCE

Day 1: Python Basics

The first day of the internship marked the beginning of my journey into the field of Artificial Intelligence and Machine Learning through Python programming. The session started with an introduction to Python, including its history, creator **Guido van Rossum**, and its importance in AI, Machine Learning, Data Science, Web Development, and Automation. The trainers explained why Python is one of the most popular programming languages due to its simplicity, readability, and extensive collection of libraries.

After the introduction, we installed the necessary software required for programming. We learned how to install and configure **Anaconda**, **Jupyter Notebook**, and **Visual Studio Code (VS Code)**. The instructor demonstrated how these tools are used in professional software development and data science projects.

Once the development environment was ready, we began learning the basics of Python programming. Topics covered included variables, data types such as integers, floating-point numbers, strings, and Boolean values. We also learned how to use arithmetic, relational, logical, and assignment operators.

The session introduced Python's **print()** and **input()** functions, allowing us to display information and receive user input. We also understood the significance of indentation in Python and how improper indentation can result in syntax errors.

By the end of the day, I became familiar with Python programming fundamentals and successfully wrote several basic programs independently.

Day 2: Control Flow and Data Structures

On the second day, we learned how Python programs make decisions and repeat tasks efficiently through control flow statements. The session began with conditional statements using **if**, **elif**, and **else**, enabling programs to perform different actions based on various conditions.

The next topic focused on loops. We learned the differences between **for loops** and **while loops**, along with their practical applications. Various examples such as multiplication tables, factorial calculations, Fibonacci series generation, star pattern printing, and number guessing games helped us understand iteration concepts effectively.

Later, we explored Python's built-in data structures.

We studied:

- Lists
- Tuples
- Sets
- Dictionaries

The day's learning helped me understand how to organize and manipulate data efficiently while writing intelligent programs.

Day 3: Functions, Modules, and Exception Handling

The third day focused on writing reusable, modular, and reliable programs.

We began by learning about **functions**, which help eliminate repeated code by grouping related statements together.

We learned how to define functions using the **def** keyword, pass arguments, return values, and create reusable blocks of code.

Several programs were developed to calculate mathematical operations, convert temperature units, determine prime numbers, and perform various utility tasks using functions.

The next topic introduced Python modules. We explored several built-in modules including:

- math
- random
- datetime

The instructor demonstrated how modules improve code organization and reduce development time by providing pre-written functionalities.

We also learned how to create custom modules and import them into different Python files, enabling better project structure and code maintenance.

Practical exercises demonstrated how exception handling prevents programs from crashing and allows developers to provide meaningful error messages to users.

By the end of Day 3, I understood the importance of modular programming and robust error handling in professional software development

Day 4: File Operations and Data Input/Output

The fourth day introduced file handling, one of the most essential concepts for data processing and machine learning.

We learned how Python interacts with external files using the **open()** function and understood different file modes such as:

- Read (“r”)
- Write (“w”)
- Append (“a”)
- Read & Write (“r+”)

Practical examples included creating text files, reading filecontents, modifying existing files, and appending new information without deleting previous data.

We also learned about the **with** statement, which automatically closes files after execution and prevents resource leakage.

The instructor introduced **CSV (Comma Separated Values)** files, which are widely used in Machine Learning datasets.

We practiced:

- Reading CSV files
- Writing CSV files
- Extracting rows and columns
- Filtering data
- Creating reports

File-related exceptions such as missing files and permission errors were also discussed, reinforcing the importance of exception handling during file operations.

By the end of the session, I understood how AI applications store, retrieve, and process large datasets using external files.

Day 5: Object-Oriented Programming (OOP) Basics

The fifth day introduced Object-Oriented Programming (OOP), a programming paradigm widely

used in software engineering and Artificial Intelligence.

The session began with understanding **classes** and **objects**. Classes act as templates for creatingobjects, while objects represent real-world entities with attributes and behaviors.

The four major principles of Object-Oriented Programming were discussed in detail:

Encapsulation

Protecting data by combining variables and methods inside a class while restricting direct access.

Inheritance

Allowing one class to inherit the properties and methods of another class, promoting code reuse and reducing redundancy.

Polymorphism

Allowing methods with the same name to behave differently depending on the object that invokes them.

Abstraction

Hiding implementation details while exposing only essential functionalities to the user. The instructor also explained how Machine Learning frameworks such as **TensorFlow**, **PyTorch**, and **Scikit-learn** internally rely heavily on Object-Oriented Programming concepts.

Day 6: Libraries for Data Science and Artificial Intelligence

The sixth day of the internship focused on learning the essential Python libraries used in Data Science, Machine Learning, and Artificial Intelligence. The session began with an introduction to **NumPy**, one of the most important libraries for numerical computing in Python.

We learned how NumPy provides multidimensional arrays that are much faster and more memory-efficient than Python lists. Topics covered included array creation, indexing, slicing, reshaping, mathematical operations, statistical functions, matrix multiplication, broadcasting, and aggregation functions. Practical exercises demonstrated how NumPy simplifies mathematical computations required in machine learning algorithms.

The next library introduced was **Pandas**, which is widely used for data manipulation and analysis.We explored two important data structures provided by Pandas:

- Series
- DataFrame

The instructor demonstrated how to load datasets from CSV files, inspect data, identify missing values, remove duplicates, sort records, filter rows based on conditions, group data, and calculate descriptive statistics. These operations showed how Pandas enables efficient handling of structured datasets.

We also discussed the importance of data preprocessing before applying machine learning algorithms. The session emphasized that well-prepared data significantly improves the performance and accuracy of predictive models.

Other useful libraries briefly introduced included:

- OpenPyXL for Excel file handling
- JSON for structured data
- Requests for interacting with APIs

By the end of Day 6, I understood how data scientists use Python libraries to efficiently collect, process, clean, and prepare data for machine learning applications

Day 7: Data Visualization and Debugging

The seventh day focused on data visualization and debugging techniques. Visualization
is one of the most important stages in data analysis because it helps identify patterns, trends, and relationships within datasets.

The session began with **Matplotlib**, Python's most widely used visualization library.

We learned how to create different types of graphs, including:

- Line Charts
- Bar Charts

- Pie Charts
- Histograms
- Scatter Plots

The instructor explained the importance of adding appropriate titles, labels, legends, and gridlines to improve graph readability.

After understanding Matplotlib, we explored **Seaborn**, a high-level visualization library built on Matplotlib. Seaborn simplifies statistical plotting and produces more attractive visualizations.

Using Seaborn, we created:

- Heat Maps
- Box Plots
- Violin Plots
- Pair Plots
- Correlation Matrices

These visualizations helped us understand relationships between variables, detect outliers, and analyze distributions in datasets.

The second half of the session focused on debugging Python programs.

We studied different types of programming errors:

- Syntax Errors
- Runtime Errors
- Logical Errors

Theinstructor demonstrated how Python displays traceback messages whenever an error occurs

We learned effective debugging techniques such as:

- Using print statements
- Reading traceback messages
- Inspecting variable values
- Using the Python Debugger (PDB)

Several debugging exercises involved correcting faulty programs, fixing logical mistakes, and improving code efficiency.

By the end of Day 7, I understood that visualization is essential for communicating insights, while debugging is a crucial skill for writing reliable software.

Day 8: Advanced Python Features

The eighth day introduced advanced programming concepts that help developers write efficient, readable, and professional-quality Python code.

We first learned about **List Comprehensions**, which provide a concise way to generate lists using loops and conditions within a single line of code.

The next topic was **Lambda Functions**, also known as anonymous functions. These are useful for writing small functions without explicitly defining them.

We used lambda functions with:

- map()
- filter()
- sorted()

The instructor then introduced **Decorators**, which allow additional functionality to be added to existing functions without modifying their code. Practical demonstrations included logging function calls and measuring execution time.

Another important concept covered was **Generators**, whichgenerate values one at a time using the **yield** keyword instead of storing the entire dataset in memory. This significantly improves memory efficiency when processing large datasets.

Finally, we learned about **Virtual Environments**, which allow developers to maintain separate project environments with different library versions. This avoids dependency conflicts and improves project organization.

By the end of Day 8, I had gained knowledge of advanced Python programming techniques that improve code efficiency, maintainability, and scalability.

Day 9: Introduction to Artificial Intelligence and Machine Learning

The ninth day marked the transition from Python programming to Artificial Intelligence and Machine Learning.

The instructor began by explaining the evolution of Artificial Intelligence and how machines can simulate human intelligence to perform tasks such as reasoning, learning, prediction, and decision-making.

We learned the difference between traditional programming and machine learning. In traditional programming:

- Data + Rules = Output

In Machine Learning:

- Data + Output = Learning Model

This fundamental difference allows computers to automatically learn patterns from historical data.

The session introduced three major categories of Machine Learning.

Supervised Learning

Supervised learning uses labeled datasets to train models.

Applications include:

- House Price Prediction
- Email Spam Detection
- Disease Prediction
- Student Performance Analysis

Unsupervised Learning

Unsupervised learning discovers hidden patterns from unlabeled datasets.

Applications include:

- Customer Segmentation
- Market Basket Analysis
- Clustering
- Recommendation Systems

Reinforcement Learning

Reinforcement Learning allows an agent to learn by interacting with an environment through rewards and penalties

Applications include:

- Self-driving Cars
- Robotics
- Game Playing
- Automated Decision Systems

The instructor explained the complete Machine Learning workflow:

- Problem Definition
- Data Collection
- Data Cleaning
- Feature Engineering

- Model Selection
- Model Training
- Model Evaluation
- Deployment
- Continuous Monitoring

Thesession concluded with a discussion on responsible AI and ethical concerns including:

- Bias in AI
- Data Privacy
- Explainability
- Fairness
- Transparency
- Accountability

By the end of Day 9, I had developed a strong understanding of Machine Learning conceptsand their real-world applications.

Day 10: Data Preparation and Feature Engineering

The tenth day focused on preparing data before applying Machine Learning algorithms.
The instructor emphasized that the quality of data directly influences model performance. The first topic covered was **Data Cleaning**.
We learned how to handle:

- Missing Values
- Duplicate Records
- Incorrect Data Types
- Formatting Errors
- Outliers

Different techniques such as removing rows, replacingmissing values using mean, median, and mode, and data normalization were demonstrated.

The next major topic was **Feature Engineering**, which involves creating meaningful input variables for machine learning models. We studied:

Encoding Categorical Variables

- Label Encoding
- One-Hot Encoding

These techniques convert categorical values into numerical form.

Feature Scaling

We learned two important scaling techniques:

- Normalization
- Standardization

Feature Creation

New features were generated by combining existing variables.

Examples included:

- Age from Date of Birth
- Day of Week from Date
- Month and Year Extraction
- Time-Based Features

These methods ensure all features have comparable ranges, improving model performance.

Text Processing

Basic Natural Language Processing techniques were introduced:

- Tokenization
- Stop Word Removal
- Stemming

- TF-IDF

Dimensionality Reduction

The instructor explained why high-dimensional datasets increase computational complexity. we learned:

- Principal Component Analysis (PCA)
- t-SNE
- UMAP

These techniques reduce the number of features while preserving important information.

By the end of Day 10, I understood that feature engineering plays a vital role in improving the accuracy and efficiency of Machine Learning models.

Day 11: Supervised Learning Algorithms

The eleventh day focused on **Supervised Machine Learning**, one of the most widely used branches of Artificial Intelligence. Supervised learning involves training a model using labeled data, where both the input and the expected output are already known. The objective is to enable the model to make accurate predictions for new, unseen data.

The first algorithm introduced was **Linear Regression**, which is used for predicting continuous numerical values. We learned how the algorithm identifies the relationship between independent variables and a dependent variable by fitting a straight line through the data points. Practical examples included predicting house prices, sales forecasting, and estimating temperatures based on historical data.

Next, we explored **Logistic Regression**, a classification algorithm used for binary prediction problems. Although its name includes “regression,” it is mainly used for classifying data into categories such as spam or not spam, pass or fail, and disease detection.

Finally, we discussed **Ensemble Learning**, which combines multiple machine learning models to improve prediction accuracy. We learned about:

- Random Forest
- Gradient Boosting
- AdaBoost

These methods reduce overfitting and increase the robustness of machine learning models.

The instructor explained evaluation metrics such as:

- Accuracy
- Precision
- Recall

These metrics help measure how effectively a model performs on unseen data.

By the end of Day 11, I gained a comprehensive understanding of supervised learning algorithms and their practical applications.

Day 12: Unsupervised Learning Algorithms

The twelfth day introduced **Unsupervised Machine Learning**, where algorithms learn hidden patterns from unlabeled datasets without predefined outputs.

The first topic covered was **Clustering**, where similar data points are grouped together automatically.

We learned the **K-Means Clustering** algorithm, which partitions data into K groups based on similarity. The instructor demonstrated how to determine the optimal number of clusters using the **Elbow Method**.

Next, we explored **Hierarchical Clustering**, which creates a hierarchy of clusters represented through dendrograms. This technique provides greater flexibility because the number of clusters does not need to be predefined.

We studied:

- Principal Component Analysis (PCA)
- t-SNE

Algorithms introduced included:

- Isolation Forest
- One-Class SVM
- Autoencoders

Applications of anomaly detection include fraud detection, cybersecurity, healthcare, and predictive maintenance.

Finally, we learned about **Association Rule Learning** using the **Apriori Algorithm**, which discovers relationships between products frequently purchased together. This technique is commonly used in recommendation systems and retail analytics.

By the end of Day 12, I understood how unsupervised learning uncovers hidden structures within data and supports intelligent decision-making.

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By the end of Day 12, I understood how unsupervised learning uncovers hidden structures within data and supports intelligent decision-making.

Day 13: Neural Networks and Deep Learning

The thirteenth day focused on **Neural Networks and Deep Learning**, one of the most advanced areas of Artificial Intelligence.

The instructor began by explaining the inspiration behind Artificial Neural Networks (ANNs),which mimic the functioning of the human brain.

A neuralnetwork consists of:

- Input Layer
- Hidden Layer(s)
- Output Layer

Each neuron receives inputs, performs computations using weights and activation functions, and passes information to the next layer.

We learned about important activation functions including:

- Sigmoid
- Tanh
- ReLU (Rectified Linear Unit)

The concept of **FeedForwardNeuralNetworks (FNNs)** was introduced, followed by practical demonstrations using **TensorFlow** and **Keras**

The session then moved to **Convolutional Neural Networks (CNNs)** , which are designed for image processing tasks such as:

- Face Recognition
- Object Detection
- Medical Image Analysis
- Autonomous Vehicles

The instructor also explained **Backpropagation**, which updates neural network weights by minimizing prediction errors using optimization algorithms such as:

- Gradient Descent
- Adam Optimizer

We concluded the day with discussions on emerging AI technologies including:

- Transformers
- Generative AI
- Large Language Models

- Transfer Learning

By the end of Day 13, I developed a strong understanding of deep learning concepts and their practical implementation.

Day 14: Model Evaluation and Optimization

The fourteenth day focused on evaluating and improving machine learning models.

The session began by discussing why evaluation is essential before deploying any machine learning model.

For classification models, we studied:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC Curve
- AUC Score

For regression models, we learned:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

We also explored **Hyperparameter Tuning** , where parameters are optimized to improve model accuracy.

Methods covered included:

- Grid Search
- Random Search

The instructor explained that selecting appropriate hyperparameters can significantly improve predictive performance.

Finally, we discussed common issues in machine learning such as:

- Overfitting
- Underfitting

By the end of Day 14, I understood how to evaluate machine learning models systematically and improve their performance through optimization techniques.

Day 15: AI Applications and Ethics

The final day focused on the real-world applications of Artificial Intelligence and the ethical responsibilities associated with developing intelligent systems The instructor discussed AI applications across various industries.

Healthcare

- Disease Diagnosis
- Medical Image Analysis
- Drug Discovery
- Personalized Treatment

Finance

- Fraud Detection
- Credit Risk Assessment
- Algorithmic Trading
- Customer Analytics

RETAIL

- Recommendation Systems
- Demand Forecasting
- Customer Segmentation
- Inventory Management

Transportation

- Autonomous Vehicles
- Traffic Prediction
- Route Optimization

Manufacturing

- Predictive Maintenance
- Quality Inspection

The second half of the session emphasized the ethical challenges of Artificial Intelligence. Important topics included:

- Algorithmic Bias
- Fairness
- Transparency
- Explainable AI
- Privacy Protection
- Accountability
- Responsible AI Development

The instructor highlighted the importance of developing AI systems that are fair, transparent, secure, and beneficial to society.

The internship concluded with an interactive discussion on future career opportunities in Artificial Intelligence, Machine Learning, Data Science, Computer Vision, Natural Language Processing, Robotics, and Generative AI.

By the end of the internship, I had developed both theoretical knowledge and practical skills that will greatly support my future career.

3.2 OBJECTIVES OF THE INTERNSHIP

The major objectives of the internship were:

- To understand the practical working environment of an IT organization.
- To enhance programming skills using Python.
- To gain knowledge of Artificial Intelligence and Machine Learning concepts.
- To learn data preprocessing and feature engineering techniques.
- To understand supervised and unsupervised learning algorithms.
- To develop problem-solving and analytical skills.
- To gain practical experience with industry-standard AI tools and libraries.
- To improve communication, teamwork, and professional ethics.
- To bridge the gap between academic learning and industrial practices.

3.3 SOFTWARE AND HARDWARE TOOLS

Software Tools

- Python 3.x
- Jupyter Notebook

- Anaconda
- Visual Studio Code
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- TensorFlow • Keras

Hardware Tools

- Laptop/Desktop Computer
- Intel Core i3/i5/i7 Processor
- Minimum 8 GB RAM
- 256 GB SSD/HDD
- Windows Operating System
- Internet Connection

CHAPTER 4
SELF ASSESMENT

4.SELF ASSESSMENT

4.1 Learning Experience and Its Outcomes

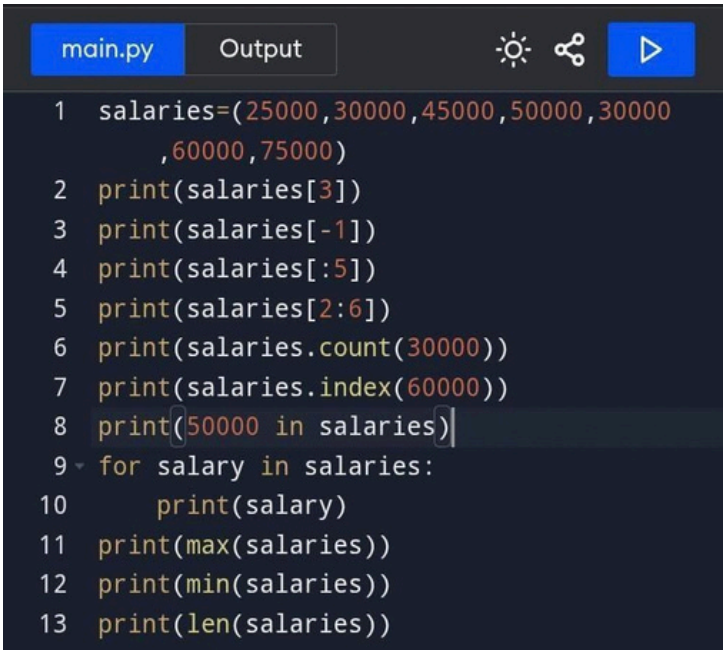
The internship at **Strydo Technologies Pvt. Ltd.** has been one of the most valuable learning experiences of my academic journey. Throughout the fifteen-day program, I gained practical exposure to Artificial Intelligence, Machine Learning, Python programming, and data science concepts that significantly enhanced my technical knowledge and confidence.

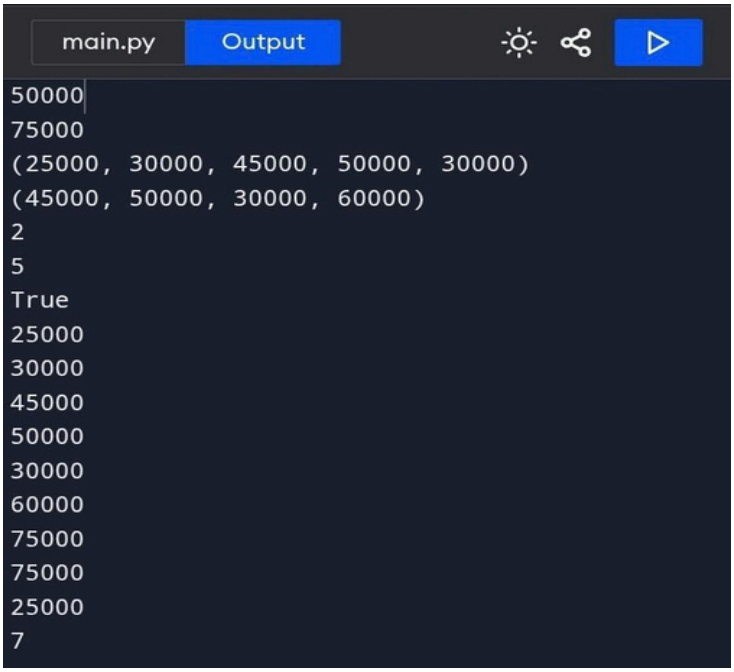
The internship strengthened my programming skills and helped me understand how theoretical concepts learned in the classroom are applied in real-world industrial environments. Working with Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, and Keras gave me hands-on experience in building and evaluating machine learning models.

Apart from technical skills, I also improved my communication skills, logical thinking, teamwork, and problem-solving abilities. The interactive sessions, practical exercises,and realworld examples encouraged me to think analytically and approach problems systematically.

Overall, this internship motivated me to continue learning advanced topics such as Deep Learning, Computer Vision, Natural Language Processing, Generative AI, and Cloud-based Machine Learning systems.

4.3 SCREENSHOT





main.py Output

```python
StudentsName=["Aaliya", "Aasifa",
    "Preethi","Hamathi","Apsana"]
print("Students Name",StudentsName[0])
print("last name",StudentsName[-1])
print("Frist 3 Students Name"
    ,StudentsName[3])
StudentsName.append("Aisha")
print("After append",StudentsName)
StudentsName.insert(2,"Aliya")
print("After insert",StudentsName)
StudentsName.extend(["Kaviya","Swetha"
    ,"hema"])
print("After extend",StudentsName)
if "Kaviya" in StudentsName:
    print("Yes Kaviya exit in the list")
else:
    print("No kaviya doesn't exist in the
        list")
if "hema" in StudentsName:
    print("hema doesn't exist on the list"
        )
else:
    print("hema exist in the list")
print("All Students Name")
```

main.py Output

```
Students Name Aaliya
last name Apsana
Frist 3 Students Name Hamathi
After append ['Aaliya', 'Aasifa', 'Preethi',
    'Hamathi', 'Apsana', 'Aisha']
After insert ['Aaliya', 'Aasifa', 'Aliya',
    'Preethi', 'Hamathi', 'Apsana', 'Aisha']
After extend ['Aaliya', 'Aasifa', 'Aliya',
    'Preethi', 'Hamathi', 'Apsana', 'Aisha',
    'Kaviya', 'Swetha', 'hema']
Yes Kaviya exit in the list
hema doesn't exist on the list
All Students Name

=== Code Execution Successful ===
```

main.py Output

```
A={10,20,30,40,50,60,50,60}
B={30,40,50,60,90}
A.add(80)
A.update([90,100])
A.remove(20)
A.discard(200)
if 50 in A:
    print("50 is exist in Set A")
else:
    print("50 doesn't exist in set A")
print ("Elements of set A:")
for i in A:
    print(i)
print("\nunion:",A.union(B))
print("intersection:",A.intersection(B))
print("Difference(A-B):",A.difference(B))
print("Symmetric Difference:",A
    .symmetric_difference(B))
print("Total elements in set A:",len(A))
```

main.py Output

```
50 is exist in Set A
Elements of set A:
10
80
90
30
100
40
50
60

union: {100, 40, 10, 80, 50, 90, 60, 30}
intersection: {40, 50, 90, 60, 30}
Difference(A-B): {80, 10, 100}
Symmetric Difference: {80, 10, 100}
Total elements in set A: 8
```

4.4 CONCLUSION

The internship at **Strydo Technologies Pvt. Ltd.** was an enriching and rewarding experience that provided me with valuable exposure to the field of Artificial Intelligence and Machine Learning. Throughout the training period, I learned Python programming, data preprocessing, machine learning algorithms, neural networks, deep learning, model evaluation, and AI ethics through both theoretical sessions and practical implementations.

The internship enhanced my technical knowledge, programming skills, analytical thinking, and confidence in solving real-world problems. It also improved my understanding of professional work culture, teamwork, communication, and continuous learning.

The knowledge and practical experience gained during this internship will serve as a strong foundation for my future academic projects and professional career in Artificial Intelligence and Machine Learning. I sincerely believe that this internship has prepared me to face future technological challenges with greater confidence and competence.

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